

function is always a straight line. In this context, *linear* means *straight line*.

When plotting the cartesian graph of a linear function, two points are sufficient to determine the line. However, in practice it is advisable to plot *three* points. If the three points lie in a straight line, it is likely that the working is correct.

Exercise 7a (Revision)

- ① Table 7.2 gives corresponding values of x and y for the equation $y = 3x - 2$:

x	-1	2	5
y	-5	4	13

Table 7.2

- Choose suitable scales and draw the cartesian graph of $y = 3x - 2$.
- Find the value of y when $x = 3.5$.
- Find the value of x when $y = 5.5$.
- Write down the coordinates of the points where the line cuts the axes.

- ② a Given that $y = 5 - 3x$, copy and complete Table 7.3.

x	-3	0	+3
y			

Table 7.3

- Hence draw the graph of the function $5 - 3x$ for values of x from -3 to +3.
 - Use the graph to solve the equation $5 - 3x = \frac{1}{2}$.
- ③ a Draw the graphs of $y = x$, $y = x + 2$, $y = 5 - x$ for values of x from 0 to +5 within the same axes.
- What do you notice about the three lines?
- ④ a Make two copies of Table 7.4.

x	-4	0	+2
y			

Table 7.4

- Complete the tables for $y = 1 - 2x$ and $y = x + 7$ and draw the graphs of these equations within the same axes.
- Use the graphs to solve the equations simultaneously.

- ⑤ Solve the following pairs of simultaneous equations graphically.

a $y = 3x$
 $y = x + 1$

b $3x - y = 12$
 $4x + 2y = 1$

(Hint: In b, first make y the subject of each equation.)

7.2 Graphs of quadratic functions

A **quadratic function** of x is an expression in x in which 2 is the highest power of x . For example, $3x^2 - 5x + 1$ is a quadratic function of x .

Example 2

Draw the cartesian graph of $y = x^2 + 2x - 3$ for values of x from -4 to +2. Use the graph to find

- the value of y when $x = 1.5$,
- the values of x when $y = 1$,
- the minimum value of $x^2 + 2x - 3$,
- the value of x when y is least.

Table 7.5 is the necessary table of values. Values of y are obtained by adding the values of x^2 , $2x$ and -3 .

x	-4	-3	-2	-1	0	1	2
x^2	16	9	4	1	0	1	4
$+2x$	-8	-6	-4	-2	0	2	4
-3	-3	-3	-3	-3	-3	-3	-3
y	5	0	-3	-4	-3	0	5

Table 7.5

Fig. 7.2 is the cartesian graph of $y = x^2 + 2x - 3$.

From the graph,

- $y = 2.25$ when $x = 1.5$
- $x = -3.25$ or 1.25 when $y = 1$
- 4 is the minimum value of $x^2 + 2x - 3$
- $x = -1$ when y has its lowest value.

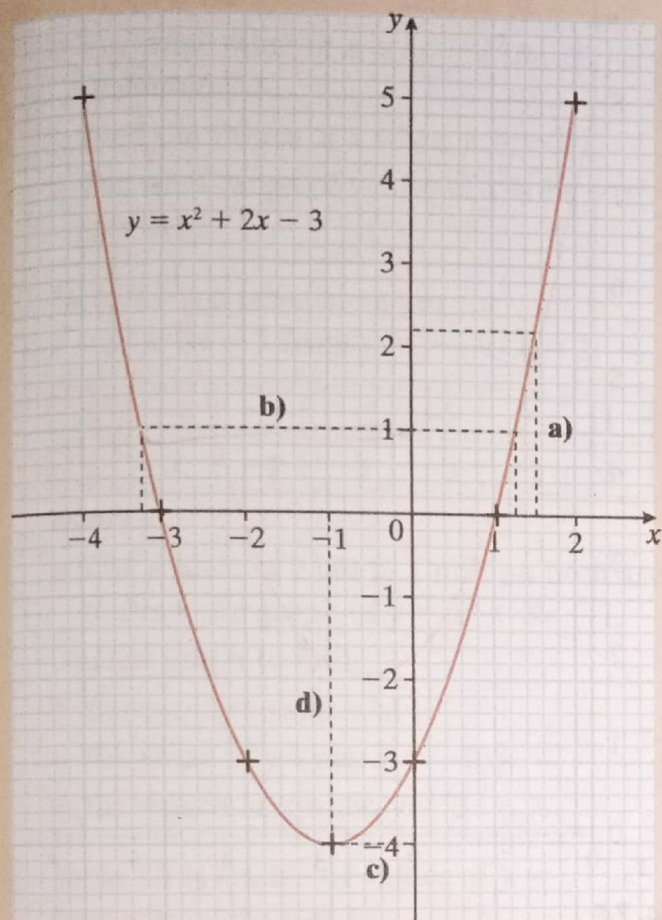


Fig. 7.2

Notice that the graph of a quadratic function is a smooth curve.

Drawing a smooth curve

It is important to be able to draw a good smooth curve. It is possible to buy special instruments that can assist this process (for example French curves; flexicurves) but these are expensive and not easily available. Similarly, perfect curves can be printed from computer-generated graphs. With practice, however, it is possible to draw a good curve *freehand*. Position your hand inside the curve and use a pencil to draw from left to right (or right to left if you are left-handed). See Fig. 7.3. In many cases this may mean turning the page around (as in Fig. 7.3).

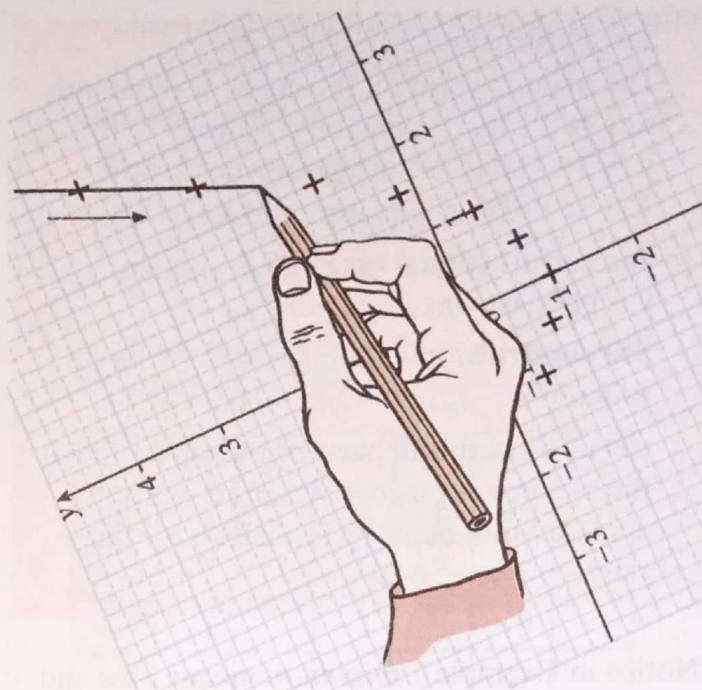


Fig. 7.3

Example 3

Fig. 7.4 is the graph of the function $2 - 3x - 2x^2$.

- What is the maximum value of $2 - 3x - 2x^2$?
- For what value of x is y greatest?
- Find the values of x when $y = -7$.
- For what range of values of x is y positive?

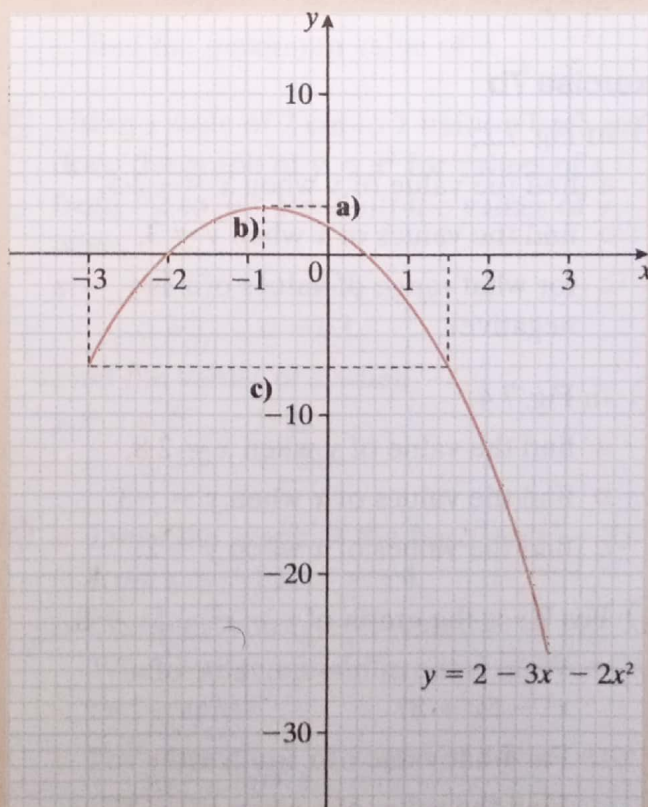


Fig. 7.4

- a** See construction **a** on Fig. 7.4.
The highest point on the curve corresponds to $y = 3.1$ (approximately). Since $y = 2 - 3x - 2x^2$, the greatest value of $2 - 3x - 2x^2$ is 3.1 (approximately).
- b** See construction **b** on Fig. 7.4.
 y is greatest when $x = -0.75$.
- c** See construction **c** on Fig. 7.4.
When $y = -7$, $x = -3$ or $+1.5$.
- d** y is positive for all parts of the curve above the x -axis. y is positive when $x > -2$ and $x < 0.5$. More neatly, y is positive for the range $-2 < x < 0.5$.

Notice in Example 2 *minimum* means *least* and in Example 3 *maximum* means *greatest*.

Some of the answers in Example 3 are only approximate. The graphs in this chapter are drawn to a small scale to save space. A bigger scale will give more accurate results.

Notice that the parabola in Fig. 7.4 is upside-down with respect to that of Fig. 7.2. When the coefficient of x^2 is positive the parabola is U-shaped; when the coefficient of x^2 is negative, the parabola is $\cap$ -shaped.

Exercise 7b

- ① In Fig. 7.2,
 - a** find the value of y when $x = -2.5$,
 - b** find the values of x when $y = 3$,
 - c** for what range of values of x is y negative?
- ② In Fig. 7.4,
 - a** find the value of y when $x = 2.8$,
 - b** find the values of x when $y = -3$
 - c** find the values of x when $y = 2$.
- ③ Fig. 7.5 is the graph of $y = x^2 - 5x + 8$.
 - a** What is the minimum value of $x^2 - 5x + 8$?
 - b** For what value of x is y least?
 - c** Find the value of y when $x = 4.5$.
 - d** Find the values of x when $y = 4.6$.

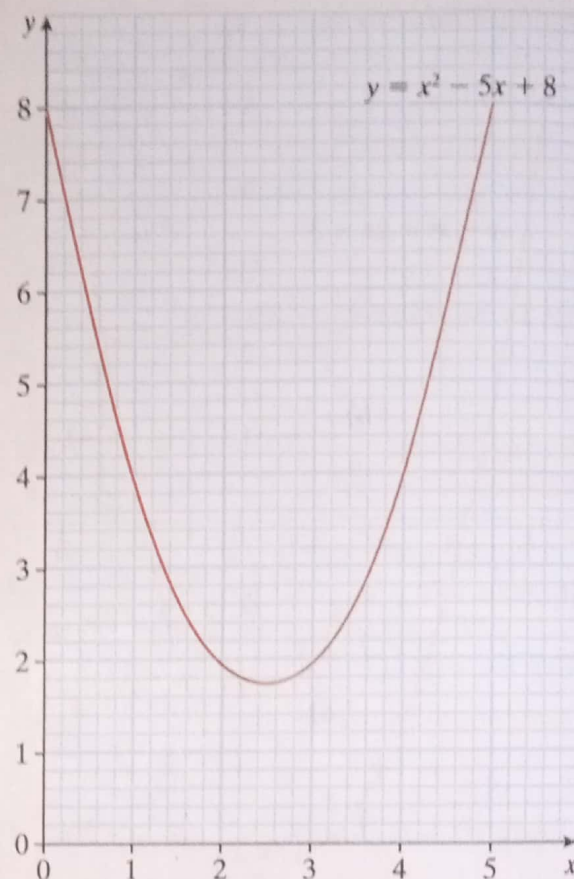


Fig. 7.5

- ④ Fig. 7.6 is the graph of the function $3x - x^2$.

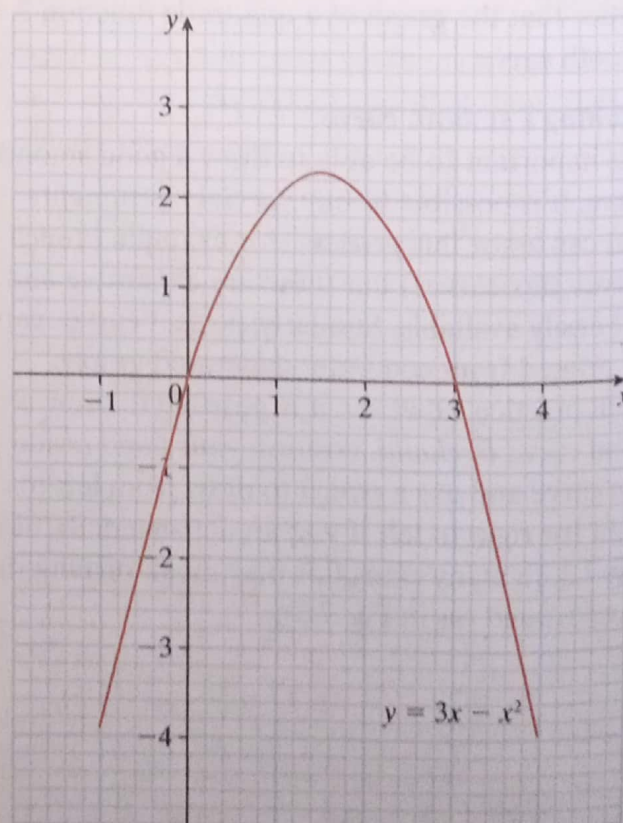


Fig. 7.6

- a What is the greatest value of y ?
- b For what value of x is $3x - x^2$ a maximum?
- c Find the values of x when $y = -3$.
- d For what range of values of x is $3x - x^2$ positive?

- 5 Table 7.6 is a table of values for $y = x^2 + 3x - 1$ from $x = -4$ to $x = 1$.

x	-4	-3	-2	-1	0	1
y	3	-1	-3	-3	-1	3

Table 7.6

- a Use a scale of 2 cm to 1 unit on both axes to draw the graph of $y = x^2 + 3x - 1$.

Use the graph to find

- b the minimum value of y ,
- c the value of y when $x = -3.5$,
- d the values of x when $y = 3$.

- 6 Table 7.7 is a table of values of $y = x - x^2$ from $x = -2$ to $x = 3$.

x	-2	-1	0	1	2	3
y	-6	-2	0	0	-2	-6

Table 7.7

Choose a suitable scale and draw the graph of $y = x - x^2$. Use the graph to

- a state the range of values of x for which y is positive,
- b find the value of x for which $x - x^2$ is a maximum,
- c find the maximum value of y ,
- d find the value of y when $x = 2.2$,
- e find the values of x when $y = -5$.

- 7 a Given that $y = x^2 - 2x - 1$, copy and complete Table 7.8.

x	-2	-1	0	1	2	3	4
x^2	4	1	0	1	4		
$-2x$	4	2	0	-2	-4		
-1	-1	-1	-1	-1	-1	-1	-1
y	7	2		-2			

Table 7.8

- b Use a scale of 2 cm to 1 unit on both axes to draw the graph of $y = x^2 - 2x - 1$.

Use your graph to find

- c the value of y when $x = 1.3$,
- d the values of x when $y = 6$,
- e the minimum value of y .

- 8 a Given that $y = 4 + 3x - x^2$, copy and complete Table 7.9.

x	-1	0	1	2	3	4
4	4	4	4	4	4	4
$+3x$	-3		3		9	
$-x^2$	-1		-1		-9	-1
y	0		6		4	

Table 7.9

- b Using a suitable scale, draw the graph of $y = 4 + 3x - x^2$.

Use your graph to

- c state the range of values of x for which $4 + 3x - x^2$ is positive,
- d find the value of y when $x = -0.6$,
- e find the values of x when $y = 3$,
- f find the maximum value of y .

- 9 Using a scale of 2 cm to 1 unit on both axes, draw the graph of $y = x^2$ for values of x from -3 to $+3$. Read off the values of y when

$x =$ a 1.5, b -0.7,
c 2.3, d -1.3.

Find the values of x when

$y =$ e 5, f 8,
g 1.9, h 0.8.

- 10 a Draw the graph of the function $x^2 - 3x$ from $x = -1$ to $x = +4$.
- b Compare your graph with that of Fig. 7.6. What is the main difference between the two graphs?
 - c For what range of values of x is $x^2 - 3x$ negative?
 - d What is the minimum value of $x^2 - 3x$?

- 11 a Draw the graph of $y = x^2 - 2x$ from $x = -2$ to $x = +4$.
- b Compare the graph with that of question 7. What is the main difference between the two graphs?
- c What is the minimum value of y ?
- d Find the values of x when $y = 7$.

- 12 a Draw the graph of $y = 5 + 4x - x^2$ from $x = -1$ to $x = +5$.

Use your graph to find

- b the value of y when $x = 1.7$,
- c the values of x when $y = 4$,
- d the range of values of x for which $y > 7$.

SUMMARY

- 1 A *linear function* of x is one that contains terms in x of power 1 only. The graph of a linear function is always a *straight line*.
- 2 A *quadratic function* of x is one in which 2 is the highest power of x . The graph of a quadratic function is a curve known as a *parabola*.
- 3 When drawing graphs, it is important to choose x - and y -axes scales carefully and to position the *origin* in a suitable place. The *range of values* of x and y will help to determine the scales and a *table of values* assists in plotting points.
- 4 When drawing a parabola, it is usually helpful to position your hand inside the curve, drawing from left to right if right-handed (or from right to left if left-handed).
- 5 When the coefficient of x^2 is positive, the parabola is U-shaped and it has a *minimum point*; when the coefficient of x^2 is negative, the parabola is n-shaped and it has a *maximum point*.

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Recall and use set notation and language
- 2 Identify the complement of a set
- 3 Write and interpret sets of values using set-builder notation
- 4 Use universal sets, complements and disjoint sets to solve practical problems
- 5 Use Venn diagrams, with up to three intersecting sets, to solve practical problems.

Teaching and learning materials

Teacher: Loops of string to make Venn diagrams; charts showing Venn diagrams.

8.1 Sets (quick revision)

Example 1

Given $\mathcal{U} = \{f, o, r, m, u, l, a, e\}$, $A = \{f, r, a, m, e\}$ and $B = \{r, u, l, e\}$,

- a draw a Venn diagram showing A , B and $\mathcal{U}$;
- b list the elements of $A \cap B$;
- c find $n(A \cup B)$.

a Fig. 8.1 is the required Venn diagram.

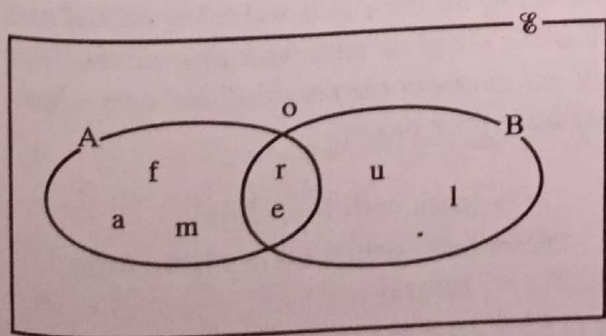


Fig. 8.1

- b $A \cap B$ is the intersection of sets A and B , i.e. the set whose elements are members of both A and B .

$$A \cap B = \{r, e\}$$

- c $A \cup B$ is the union of sets A and B , i.e. the set whose elements are members of A or B or both A and B .

$$A \cup B = \{f, r, a, m, e, u, l\}$$

Notice that although r and e are elements of both A and B , there is no need to write them down twice.

$n(A \cup B)$ is the number of elements in the set $A \cup B$.

$$n(A \cup B) = 7$$

Table 8.1 contains some of the symbols and language of sets that appeared in Chapter 5.

symbols	meaning
$P = \{a, b, c, d\}$	P is the set a, b, c, d
$Q = \{1, 2, 3, \dots\}$	$\dots$ means 'and so on'
$\emptyset$ or $\{\}$	the empty set
$\mathcal{U}$ or $\mathcal{U}$	the universal set
$\in$	is a member of
$\notin$	is <i>not</i> a member of
$\subset$	is a subset of
$\not\subset$	is <i>not</i> a subset of
$\supset$	contains
$P \cup Q$	union of P and Q
$P \cap Q$	intersection of P and Q
$n(S)$	number of elements in S

Table 8.1

Exercise 8a (Revision)

- ① Make each of the following true by writing either $\in$ or $\notin$ in place of the $*$.
 - a $9 * \{2, 4, 6, 8, 10\}$
 - b $15 * \{3, 6, 9, \dots, 24\}$
 - c $44 * \{5, 10, 15, 20, \dots\}$
 - d $R * \{a, b, c, d, \dots, z\}$

② Find $n(X)$ when $X =$

- a** {h, o, u, s, e} **b** {toes on your feet}
c {0, 1, 2} **d** {months in a year}
e {5, 5, 6, 6} **f** {11, 12, 13, ..., 22}

③ Give three examples of an empty set.

④ Give three examples of an infinite set.

⑤ Write down all the subsets of the following.

- a** {3, 4, 5} **b** {x, y}
c {0, 2} **d** {f, o, u, r}

⑥ Write down the following using symbols.

- a** 2 and 6 are a subset of the factors of 18
b {trees} is not a subset of {metal objects}
c {vehicles} contains {buses}
d children are members of the human race

⑦ If $\mathcal{E} = \{\text{months of the year}\}$, $F = \{\text{first eight months of the year}\}$, $Y = \{\text{months ending in y}\}$, draw a Venn diagram to show the relationship between F , Y and $\mathcal{E}$.

⑧ Which of the following pairs of sets are disjoint? If the sets are not disjoint, write down two members of the intersection.

- a** {prime factors of 24}, {prime factors of 55}
b {multiples of 5}, {multiples of 7}
c {Zimbabwe, Zaria, Zambia}, {countries of Africa}
d {letters of *bull*}, {letters of *cow*}

⑨ If $\mathcal{U} = \{2, 4, 6, 8, \dots, 20\}$

$M = \{\text{multiples of 6 less than 20}\}$

$L = \{\text{even numbers less than 14}\}$

write down the members of the following sets.

- a** $M \cup L$ **b** $M \cap L$ **c** $M \cup \mathcal{U}$
d $\mathcal{U} \cap M$ **e** $\mathcal{U} \cap L$ **f** $L \cup \mathcal{U}$

⑩ **a** Draw a Venn diagram to represent the data of question 9.

b Hence find

- i** $n(M \cup L)$, **ii** $n(M \cap L)$.

8.2 Complement of a set

In Fig. 8.2, A is a subset of the universal set $\mathcal{E}$, i.e. $A \subset \mathcal{E}$.

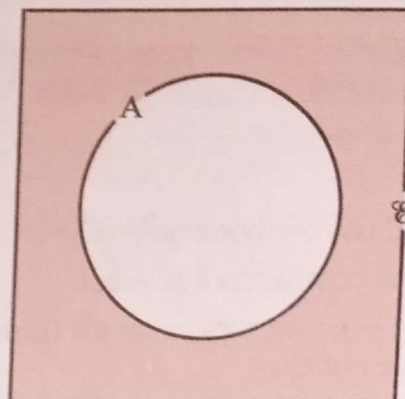


Fig. 8.2

The **complement** of A is the set which contains all those elements of $\mathcal{E}$ which are *not* members of A . A' is short for the complement of A . In Fig. 8.2 the shaded region represents A' .

Example 2

If $\mathcal{E} = \{1, 2, 3, 4, 5\}$, $A = \{1, 3\}$ and $B = \{3, 4\}$ find

a A' , **b** B' , **c** $(A \cap B)'$, **d** $(A \cup B)'$.

a $A' = \{2, 4, 5\}$

b $B' = \{1, 2, 5\}$

c $A \cap B = \{3\}$
 $(A \cap B)' = \{1, 2, 4, 5\}$

d $A \cup B = \{1, 3, 4\}$
 $(A \cup B)' = \{2, 5\}$

Example 3

$\mathcal{E}$ is the set of all cars, R is the set of all red cars and F is the set of all cars with four doors. Show on a Venn diagram the set of all red cars which do not have four doors.

$F = \{\text{cars with four doors}\}$

$F' = \{\text{cars which do not have four doors}\}$

$R \cap F' = \{\text{red cars which do not have four doors}\}$

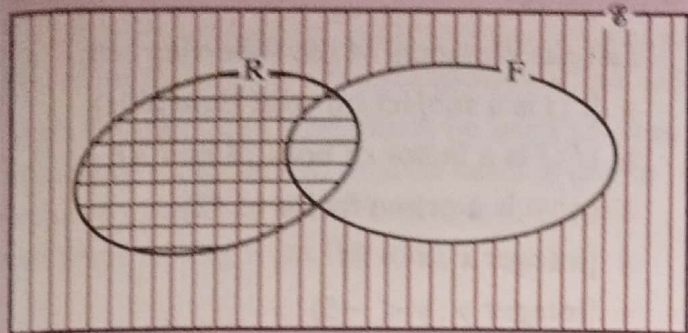


Fig. 8.3

In Fig. 8.3, the region with horizontal shading represents R . The region with vertical shading represents F . The region with cross-shading represents $R \cap F$, the set of all red cars which do not have four doors.

Exercise 8b

- ① Given $\mathcal{U} = \{1, 2, 3, 4, 6, 8\}$, write down the complements of the following sets.

a $\{1, 2, 3\}$	b $\{4, 6, 8\}$
c $\{2, 4\}$	d $\{1, 3, 6, 8\}$
e $\{6\}$	f $\{1, 2, 3, 4, 6\}$
g $\emptyset$	h $\mathcal{U}$
i $\{1, 2, 3, 6\} \cup \{1, 2, 8\}$	
j $\{1, 2, 3, 6\} \cap \{1, 2, 8\}$	
- ② If $\mathcal{U} = \{a, b, c, d, e\}$, $A = \{a, c, e\}$ and $B = \{b, e\}$, list the members of the following sets.

a A'	b B'	c $A' \cup B$
d $A \cup B'$	e $A' \cap B$	f $A \cap B'$
g $A' \cup A$	h $B' \cap B$	i $(A \cup B)'$
j $A' \cup B'$	k $(A \cap B)'$	l $A' \cap B'$
- ③ Compare the answers to parts i and l of question 2. What do you notice?
- ④ Compare the answers to parts j and k of question 2. What do you notice?
- ⑤ If $\mathcal{U} = \{1, 2, 3, 4, 5, \dots, 10\}$, $A = \{1, 2, 5, 7\}$, $B = \{1, 3, 6, 7\}$, write down the sets A' , B' , $(A \cap B)'$ and $(A \cup B)'$.
- ⑥ Using the sets of question 5, show that $A' \cup B' = (A \cap B)'$ and $A' \cap B' = (A \cup B)'$.
- ⑦ Use Venn diagrams to illustrate the results of question 6.

- ⑧ If $\mathcal{U} = \{\text{days of the week}\}$, $S = \{\text{days which contain the letter s}\}$ and $N = \{\text{days which contain six letters}\}$
 - a list the members of the sets S , N , S' , N' ,
 - b list the members of the set
 - i $(S \cup N)'$, ii $(S \cap N)'$,
 - c hence, without further working, list the members of the sets
 - i $S' \cap N'$, ii $S' \cup N'$.
- ⑨ Use a Venn diagram to show that if $A \subset B$ then $B' \subset A'$, and vice versa.
- ⑩ If $\mathcal{U} = \{\text{all teachers}\}$, $M = \{\text{mathematics teachers}\}$ and $W = \{\text{teachers who are women}\}$, show on a Venn diagram the set of all mathematics teachers who are men.

8.3 Set-builder notation

$A = \{\text{integer } x : x < 4\}$ is short for 'A is the set of integers, x , such that x is less than 4'. This is an example of **set-builder notation**. In set-builder notation, the colon ($:$) stands for 'such that'. If set-builder notation is expressed properly, it is always possible to list the elements of the given set. For example, in this case

$$A = \{\dots, -2, -1, 0, 1, 2, 3\}$$

Notice that the elements of A are restricted to integral or whole-number values. The letter $\mathbb{Z}$, sometimes written $\mathbb{Z}$, is often reserved for the set of integers:

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, +1, +2, +3, \dots\}$$

Hence set A can be rewritten

$$A = \{x : x < 4, x \in \mathbb{Z}\}$$

i.e. 'A is the set of values of x such that x is less than 4, where x is an integer'.

If, however, the elements of A are restricted to natural numbers, i.e. positive whole-numbers, then

$$A = \{1, 2, 3\}$$

The letter $\mathbb{N}$, sometimes written $\mathbb{N}$, is often reserved for the set of natural numbers:

$$\mathbb{N} = \{1, 2, 3, \dots\} \text{ or } \mathbb{N} = \{1, 2, 3, \dots\}$$

Hence set A can be rewritten:

$$A = \{x : x < 4, x \in \mathbb{N}\}$$

i.e. 'A is the set of values of x such that x is less than 4, where x is a natural number'.

Example 4

List the members of the set $\{x : x \geq -2, x \in \mathbb{Z}\}$.

$$\{x : x \geq -2, x \in \mathbb{Z}\} = \{-2, -1, 0, 1, \dots\}$$

Remember the meanings of the following symbols:

$>$ is greater than

$\geq$ is greater than or equal to

$<$ is less than

$\leq$ is less than or equal to

Example 5

Given that x is an integer, list the members of $\{x : x < 4\} \cap \{x : x \geq -2\}$.

If $x \in \mathbb{Z}$,

$$\{x : x < 4\} = \{\dots, -3, -2, -1, 0, 1, 2, 3\}$$

$$\{x : x \geq -2\} = \{-2, -1, 0, 1, 2, 3, 4, \dots\}$$

$$\{x : x < 4\} \cap \{x : x \geq -2\} = \{-2, -1, 0, 1, 2, 3\}$$

Notice, in Example 5, that the expression $\{x : x < 4\} \cap \{x : x \geq -2\}$ can be shortened to $\{x : -2 \leq x < 4\}$, i.e. 'the set of values, x , such that x is greater than or equal to -2 and less than 4 '. The expression $-2 \leq x < 4$ is a **range of values** of x .

Example 6

List the members of the set $\{y : 1 < y \leq 6, y \in \mathbb{N}\}$.

$$\{y : 1 < y \leq 6, y \in \mathbb{N}\} = \{2, 3, 4, 5, 6\}$$

Example 7

If $A = \{1, 2, 3, 4\}$, rewrite A using set-builder notation.

There are many possible solutions. For example

$$\text{Either } A = \{x : 1 \leq x \leq 4, x \in \mathbb{N}\}$$

$$\text{or } A = \{x : 0 < x < 5, x \in \mathbb{Z}\}$$

$$\text{or } A = \{x : x < 5, x \in \mathbb{N}\}$$

The letter $\mathbb{Q}$, sometimes written $\mathbb{Q}$, is often reserved for the set of rational numbers.

$$\mathbb{Q} = \{\text{numbers of the form } \frac{a}{b} \\ \text{where } a \in \mathbb{Z} \text{ and } b \in \mathbb{N}\}$$

$$\mathbb{Q} = \{\frac{a}{b} : a \in \mathbb{Z}, b \in \mathbb{N}\}$$

Exercise 8c

① List the elements of the following sets.

a $\{t : t \text{ is a subject on your timetable}\}$

b $\{f : f \text{ is a factor of both 24 and 30}\}$

c $\{p : p \text{ is a prime factor of 42}\}$

d $\{\text{integer } x : x > 8\}$

e $\{\text{integer } y : y < -5\}$

f $\{x : 3 < x < 5, x \in \mathbb{N}\}$

g $\{x : 3 < x \leq 5, x \in \mathbb{N}\}$

h $\{x : -2 < x < -1, x \in \mathbb{Z}\}$

② If $\mathbb{E} = \{\text{positive integers}\}$, list the members of the following sets.

Note: 0 (zero) is neither positive nor negative.

a $\{x : x > 7\}$

b $\{x : x > -3\}$

c $\{x : x < -1\}$

d $\{x : x < 10\}$

e $\{x : x - 3 = 0\}$

f $\{x : x + 3 = 9\}$

g $\{x : x + 2 = 0\}$

h $\{x : 2x = 7\}$

③ Assuming that x is an integer, state the lowest and highest values of x for each of the following ranges.

a $-2 < x < 5$

b $-8 \leq x \leq 9$

c $-5 < x \leq 0$

d $2 \leq x < 10$

e $4 < x < 6$

f $-8 < x < -1$

g $-7 \leq x < 1$

h $-5\frac{2}{3} < x < 1\frac{3}{4}$

④ Using the inequality symbol $<$, write the following in set-builder notation.

a $\{0, 1, 2, 3\}$

b $\{-1, 0, 1\}$

c $\{-3, -2, -1, 0\}$

d $\{-9, -8, \dots, 3, 4\}$

e $\{9, 10\}$

f $\{-8, -7, \dots, -1\}$

⑤ If $x \in \mathcal{U}$ and $\mathcal{U} = \{1, 2, 3, \dots, 10\}$, list the members of the following subsets of $\mathcal{U}$.

a $\{x : x \text{ is an odd number}\}$

b $\{x : x < 7\}$

c $\{x : x + 2 \geq 7\}$

d $\{x : x - 2 = 2\}$

e $\{x : 3x - 2 = 19\}$

f $\{x : x \text{ is a factor of 24}\}$

8.4 Venn diagrams and problem solving

Venn diagrams can sometimes be used to store numerical information. In such cases it is also possible to use the Venn diagrams to solve problems arising from the data.

Example 8

50 students were asked what they did last night. 16 said they read a book, 41 said they watched television. If 7 said they did neither, how many did both?

Let $\mathcal{E} = \{\text{all students}\}$, $B = \{\text{book readers}\}$ and $T = \{\text{television watchers}\}$. It is required to find $n(B \cap T)$. Let $n(B \cap T) = x$.

Fig. 8.4 is a Venn diagram containing the given information. The numbers in the regions of the Venn diagram represent the numbers of elements in the regions.

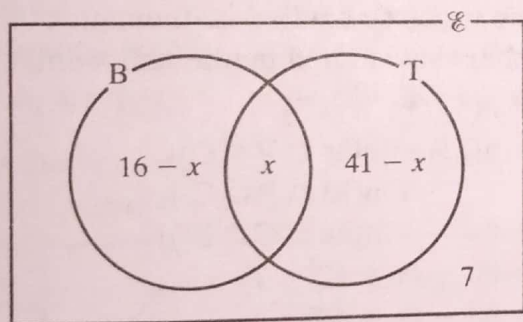


Fig. 8.4

$$\begin{aligned}\text{Since } n(T) &= 41, n(B' \cap T) \\ &= 41 - x.\end{aligned}$$

$$\begin{aligned}\text{Similarly, } n(B \cap T') &= 16 - x \\ \text{and } n(B \cup T)' &= 7\end{aligned}$$

The totals for the regions must add up to the number of people in the universal set:

$$\begin{aligned}x + (16 - x) + (41 - x) + 7 &= 50 \\ \Leftrightarrow 64 - x &= 50 \\ \Leftrightarrow x &= 14\end{aligned}$$

14 students read a book *and* watched television.

Note: The 16 who read a book includes the 14 who also watched television.

Example 9

In a survey of students, it was discovered that 76 like history, 98 like geography and 87 like mathematics. 15 like all the three subjects, 39 like history and geography, 37 like history and mathematics, 48 like geography and mathematics and 45 like none of the subjects.

- Show this information using a Venn diagram.
- How many students took part in the survey?

Let $\mathcal{E} = \{\text{all students that took part in the survey}\}$, $G = \{\text{those who like geography}\}$, $H = \{\text{those who like history}\}$, $M = \{\text{those who like mathematics}\}$.

- Fig. 8.5 is a Venn diagram containing the given information. The numbers in the regions represent the numbers of students in each region. (Check that the data in Fig. 8.5 agree with those in the question.)

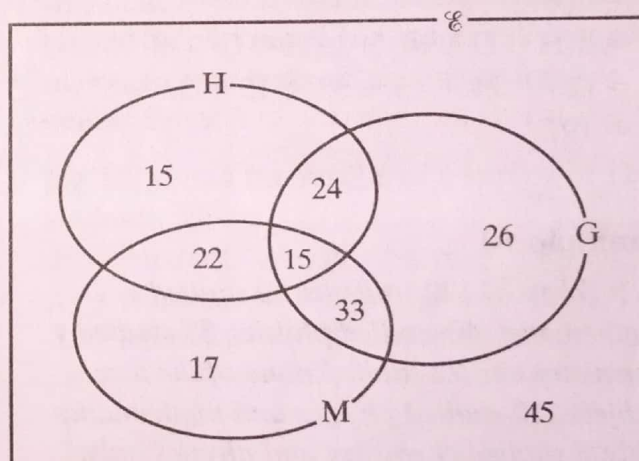


Fig. 8.5

Read the following carefully. It shows how the numbers in the regions of Fig. 8.5 were obtained.

$$\begin{aligned}\text{Since } n(H \cap G \cap M) &= 15 \text{ and} \\ n(H \cap G) &= 39 \\ n(H \cap G \cap M') &= 39 - 15 \\ &= 24\end{aligned}$$

Similarly

$$\begin{aligned}n(H \cap M \cap G') &= 37 - 15 \\ &= 22\end{aligned}$$

and

$$\begin{aligned}n(G \cap M \cap H') &= 48 - 15 \\ &= 33\end{aligned}$$

The number of students who like history only

$$\begin{aligned} &= n(H) - 15 - 22 - 24 \\ &= 76 - 61 \\ &= 15 \end{aligned}$$

Similarly, the number of students who like geography only

$$\begin{aligned} &= n(G) - 15 - 24 - 33 \\ &= 98 - 72 \\ &= 26 \end{aligned}$$

And the number of students who like mathematics only

$$\begin{aligned} &= n(M) - 15 - 33 - 22 \\ &= 87 - 70 \\ &= 17 \end{aligned}$$

- b** Number of students that took part in the survey
- $$\begin{aligned} &= n(\mathcal{U}) \\ &= n(H \cup G \cup M) + n(H \cup G \cup M)' \\ &= (26 + 24 + 15 + 33 + 15 + 22 + 17) + 45 \\ &= 197 \end{aligned}$$

Example 10

In a group of 120 students, 60 studied mathematics, 40 studied physics, 55 studied chemistry and 22 studied none of the three subjects. 12 studied physics and mathematics only, 8 studied chemistry and physics only, 7 studied mathematics and chemistry only.

- a** Draw a Venn diagram illustrating the information.
- b** Find the number of students that studied all three subjects.
- c** Find the number of students that studied only mathematics.

Let $\mathcal{U}$ = {all the students in the class},
 M = {those that studied mathematics},
 P = {those that studied physics},
 C = {those that studied chemistry}.

Let $n(M \cap P \cap C) = x$.

- a** Fig. 8.6 is a Venn diagram containing the given information. Check Fig. 8.6 carefully against the data.

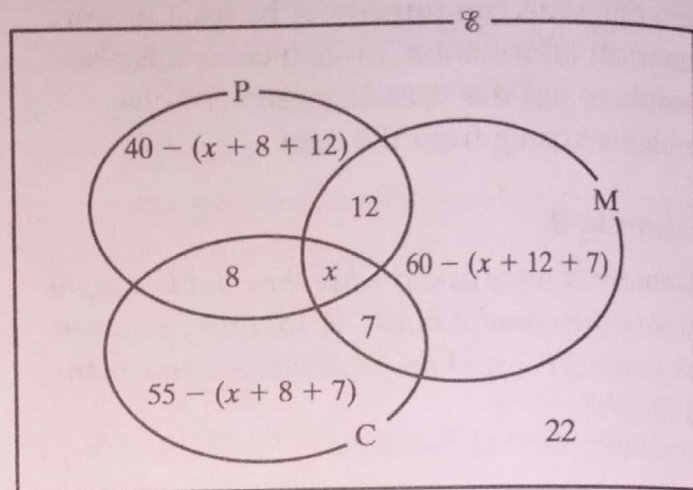


Fig. 8.6

Since 12 studied physics and mathematics only,

$$n(P \cap M \cap C') = 12.$$

Similarly, $n(C \cap P \cap M') = 8$

and $n(M \cap C \cap P') = 7$.

Again, since 60 studied mathematics, the number that studied mathematics only

$$\begin{aligned} &= n(M \cap P' \cap C') \\ &= n(M) - [n(M \cap P \cap C) \\ &\quad + n(M \cap P \cap C') \\ &\quad + n(M \cap C \cap P')] \\ &= 60 - (x + 12 + 7) \\ &= 60 - (x + 19) \end{aligned}$$

Similarly $n(P \cap M' \cap C') = 40 - (20 + x)$
 and $n(C \cap M' \cap P') = 55 - (15 + x)$.

- b** The total for the regions must add up to the number of students in the class.

$$\begin{aligned} &40 - (20 + x) + 60 - (19 + x) + 55 - (15 + x) + 12 + 8 + 7 + x + 22 = 120 \\ \Leftrightarrow &155 - (54 + 3x) + 49 + x = 120 \\ \Leftrightarrow &150 - 2x = 120 \\ \Leftrightarrow &x = 15 \end{aligned}$$

15 studied all three subjects.

- c** The number that studied only mathematics
- $$\begin{aligned} &= n(M \cap P' \cap C') \\ &= 60 - (19 + 15) \\ &= 26. \end{aligned}$$